

VISHAL DHAWAL

+91 7880368897 | vishaldhawal8853@gmail.com | [LinkedIn](#) | [Github](#) | [Medium](#)

Computer Science undergraduate specializing in Computer Vision and Deep Learning with experience spanning aerial image analysis, threat intelligence, and vision-based Human-Computer Interaction, using PyTorch, OpenCV, EfficientNet, and MediaPipe.

EDUCATION

- Indian Institute of Information Technology Kottayam, India**

Bachelor of Technology - Computer Science and Engineering;

GPA: 7.3/10 (through Sem 6)

Aug 2023 - April 2027

- Indian Institute of Technology Mandi, India**

Minor in Computer Science and Advanced Technologies;

GPA: 7.4/10

Oct 2024 - Oct 2025

PROJECTS

- Aerial GCP Pose** Computer Vision • PyTorch • EfficientNet B3 • Multi-Task learning

June 2026, [Github](#)

- Built a multi-task EfficientNet-B3 pipeline for GCP localization and shape classification from 4096×3000 aerial imagery using crop-based training and keypoint-aware augmentation.
- Designed a leakage-safe training and tiled inference pipeline with 512×512 crops and GroupKFold grouping by project/survey/GCP.
- Achieved 20.32 px mean localization error, ~100% shape classification accuracy, and ~1.0 Macro-F1, with inference successfully executed across 300 unlabeled test images.

- VisionControl** Human Computer Interaction • OpenCV • MediaPipe • PyAutoGUI

December 2025, [Github](#)

- Implemented a vision-based gesture system using OpenCV and MediaPipe to smoothly control cursor, clicks, volume, and brightness in real time with high accuracy.
- Achieved 92% gesture-tracking accuracy and 85% blink-click reliability by optimizing hand + face landmark processing.
- Improved interaction stability by 40% using smoothing + motion-filtering, resulting in smoother and more responsive control.

- TeleWire** Telegram SOCMINT • Threat Intelligence • NLP • Telethon

July 2026, [Github](#)

- Developed an AI-powered Telegram threat intelligence platform for automated monitoring, entity extraction, semantic threat classification, and cybercrime intelligence generation across public channels.
- Automated extraction of entities, semantic threat classification, OCR, and composite risk scoring to assist intelligence-driven cyber investigations.
- Built interactive investigation dashboards with campaign clustering, relationship visualization, and case reporting to support efficient analyst decision-making.

EXPERIENCE

- AI/ML Intern — Cyber Crime Headquarters, Uttar Pradesh Police**

(June 2026 - July 2026)

- Developed CYBORG, an AI-powered investigation assistant leveraging LLMs, semantic retrieval, and document intelligence to assist cybercrime investigations through intelligent knowledge retrieval and contextual analysis.
- Designed AI workflows for knowledge retrieval and decision support while collaborating with senior law enforcement officials under strict operational confidentiality.

- Mentor, MindQuest Mental Wellness Club – IIIT Kottayam**

(Sept 2025 – Aug 2026)

- Mentored volunteers through interviews, the GameOn event (300+ participants), and student engagement initiatives
- Contributed to planning, organizing, and executing mental wellness programs to strengthen student outreach and participation.

TECHNICAL SKILLS

- Programming:** Python, C, C++, SQL
- Machine Learning & AI:** CNNs, NLP, Transformers (BERT), LLMs, Retrieval-Augmented Generation (RAG)
- Computer Vision:** Hand Tracking, Face Mesh, Eye Tracking, Gesture Recognition, OpenCV, MediaPipe
- Tools & Platforms:** Git & GitHub, FastAPI, WebSocket, VS Code, Jupyter Notebook, Google Colab, Firebase
- Data Analytics:** Pandas, NumPy, Matplotlib, Data Visualization, Statistical Analysis, EDA
- Interests:** Computer Vision, Human Computer Interaction, Applied Machine Learning, Hyperspectral Image Processing

CERTIFICATION

- OpenCV BootCamp
- ChatGPT Prompt Engineering for Developers

March 2025, [Certificates](#)

March 2025, [Certificates](#)